Super keyword & constructor chaining in Java

Definition

- The super keyword in Java is a reference variable that refers to the immediate parent (superclass) object. It is used to access the parent class's variables, methods, and constructors from the child class.

Uses of super Keyword

- There are three main uses of the super keyword:
- Access the parent class variable
- Call the parent class method
- Call the parent class constructor

1. Access Parent Class Variable

When both parent and child classes have variables with the same name,**super** is used to access the parent class variable

```
class Animal {
    String color = "White";
}

class Dog extends Animal {
    String color = "Black";

    void display() {
        System.out.println("Dog Color : " + color);
        System.out.println("Animal Color : " + super.color);
    }
}

public class SuperVariableDemo {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.display();
    }
}
```

Output

```
Dog Color : Black
Animal Color : White
```

2. Call Parent Class Method

If the child class overrides a method, `super` can call the parent class version.

```
class Animal
{
    void sound()
{
        System.out.println("Animal makes a sound");
    }
}

class Dog extends Animal
{

    void sound()
{
        super.sound();
        System.out.println("Dog barks");
    }
}

public class Demo
 {
    public static void main(String args[])
{
        Dog d = new Dog();
        d.sound();
    }
}
```

Output

```
Animal makes a sound
Dog barks
```

3. Call Parent Class Constructor

super() is used to invoke the constructor of the parent class

```
class Animal
{

    Animal()
 {
        System.out.println("Animal Constructor");
    }
}

class Dog extends Animal
 {

    Dog()
{
        super();
        System.out.println("Dog Constructor");
    }
}

public class Demo
{
    public static void main(String args[])
{
        Dog d = new Dog();
    }
}
```

Output

```
Animal Constructor
Dog Constructor
```

Constructor Chaining in Java

Definition

Constructor chaining is the process of calling one constructor from another constructor. It helps avoid duplicate code and ensures constructors are executed in a proper sequence.

Constructor chaining can happen in two ways:

Within the same class using this()
Between parent and child classes using super()

1. Constructor Chaining Using `this()`

The `this()` keyword is used to call another constructor of the **same class.**

```java
class Student {

    Student() {
        this(101);
        System.out.println("Default Constructor");
    }

    Student(int id) {
        System.out.println("Student ID: " + id);
    }
}

public class ThisChainingDemo {
    public static void main(String[] args) {
        Student s = new Student();
    }
}
```

Output

```
Student ID: 101
Default Constructor
```